

Asher Norland

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EXPERIENCE

Firaxis Games

Graphics Programmer

Civilization VII

- ▶ Created core procedural systems for in-game environments, including vegetation placement, buildings distribution, and river construction.
 - ▶ Implemented parametric curve generation and sampling within a mathematically continuous terrain generation process
- ▶ Worked in close collaboration with multidisciplinary teams of artists and programmers, ensuring alignment and integration of product features across the product's lifecycle.
 - ▶ Designed and built procedural modifiers to streamline procedural asset hierarchies within the asset editor, enabling extensive asset re-use and asset prototyping
 - ▶ Integrated an artist-friendly scripting grammar with the engine, empowering artists to control procedural asset creation and behavior
- ▶ Implemented efficient CPU ray tracing algorithms for both single and batched intersection queries.
 - ▶ Developed generic and specialized acceleration structures to enable rapid scene updates, improving performance of procedural systems, game units, and visual effects.
- ▶ Managed asset performance via extensive and automated performance testing.
 - ▶ Designed and implemented the player-facing graphics and AI benchmarking, providing valuable comparison data for both developers and end users.
 - ▶ Automated an asset benchmarking process through scriptable tools, facilitating internal performance evaluations and providing artists with quantitative feedback.

Marvel's Midnight Suns

- ▶ Provided launch support for HDR rendering.

Disbelief

Programmer I

Cambridge, MA

Aug. 2018 – Sept. 2020

- ▶ Provided specialist services to clients in the games and hardware industry within their development workflow.
- ▶ Audited Unreal Engine and other software for their applicability to modern consoles.
- ▶ Contributed to major AR, VR, and console platforms, including their SDK and subsequent Unreal Engine integrations.
- ▶ Optimized, debugged, and created features for the Unreal Engine, including audio, graphical post processing, and the engine's use of the Android platform.

Wolves in the Walls

- ▶ Made UE4 engine graphics pipeline changes to support global buffers for stylized rendering
- ▶ Addressed performance issues within stylized fragment shaders based on published research

RPI – Center for Architecture Science and Ecology

Undergraduate Research Assistant

Troy, NY

Jan. 2017 – Jan. 2018

- ▶ Developed web frontend and backend for data visualization funded by the Bill & Melinda Gates Foundation.
- ▶ Integrated WebGL support for three-dimensional mesh manipulation to enable architectural visualization.

EDUCATION

Rensselaer Polytechnic Institute (RPI)

B.S in Computer Science

Troy, NY

Aug. 2014 – May 2018

SKILLS & INTERESTS

Programming Languages: *Proficient* C++, C, Python, CMake, Shading Languages; *Familiar* TypeScript, C#

Technologies: *Proficient* CUDA, Vulkan, DirectX 12, Git, Perforce; *Familiar* MPI, OpenCL

Software: *Proficient* Windows, Nsight, RenderDoc, Unreal Engine, AR/VR platforms; *Familiar* Linux, Switch

Interests: Film, Hiking, Photography, Rock Climbing